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SIZING UP ENTREPRENEURIAL POTENTIAL: GENDER DIFFERENCES IN COMMUNICATION AND INVESTOR PERCEPTIONS OF LONG-TERM GROWTH AND SCALABILITY

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Female entrepreneurs have been found to face disadvantages when compared with male entrepreneurs, especially in acquiring the financial resources needed to sustain and grow their ventures. Across three studies, we examine how disparities in funding outcomes may be due to differences in how entrepreneurs communicate their ventures, whereby female entrepreneurs have a tendency to use more concrete language when describing their ventures than their male counterparts. We find that the use of abstract speech affects investors' perceptions of which ventures are oriented toward long-term growth and scalability, which in turn affects the likelihood that a venture will receive investment. We conclude with a discussion of the important role of communication style as a key mediating mechanism in influencing investor cognition.

> I see men pitch unicorns, and women pitch businesses.
>
> —Venture Capitalist

Entrepreneurs who start their own ventures do so knowing that they face extremely high failure rates (Aldrich & Ruef, 2006). To increase their odds of success, entrepreneurs attempt to mobilize key resources, particularly financial capital, from early-stage investors (Clough, Fang, Vissa, & Wu, 2019). Recent research has shown that female entrepreneurs face significant disadvantages in securing the necessary financial resources to support their ventures (Brooks, Huang, Kearney, & Murray, 2014). Several reasons have been given to explain this disparity. Some have claimed that investors exhibit bias against female entrepreneurs, (implicitly) deeming them as lacking in the qualities necessary to engage in successful early-stage venturing (Lee & Huang, 2018). Others have argued that gender disparities in funding outcomes are not due to any pernicious discrimination, but that investors provide women with a smaller share of funds simply because female entrepreneurs seek, and require, less funding for their ventures in the first place (Brush, Greene, Balachandra, & Davis, 2018).

While these explanations may partially explain the disparities in financing outcomes, behavioral tendencies on the part of the entrepreneur—i.e., the style and the manner in which entrepreneurs present and convey information about their ventures in the first place—are also likely to play a key role. Studies have shown that there may be differences in how women and men generally communicate (Burgoon, Dillard, & Doran, 1983; Lakoff, 1975), suggesting that

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the communicative styles used by male and female entrepreneurs in presenting their ventures may also differ. More specifically, recent research has demonstrated that women tend to speak about specifics, whereas men tend to speak about the bigger picture (Joshi, Wakslak, Appel, & Huang, 2020). As our opening quote suggests, investors have anecdotally described a similar pattern within the entrepreneurial context, noting how women often "pitch the business they're building right now" (Bellstrom, 2017), unlike men, who tend to describe a larger vision for how they see their business growing and expanding over time.

Therefore, in our current investigation, we draw on prior research on gendered communication, construal level theory, and entrepreneurial decision making to argue that male and female entrepreneurs differ in the degree to which they describe their ventures abstractly (i.e., with a focus on the bigger picture) versus concretely (i.e., in a more contextualized and specific manner), and that this difference in communicative style is an important driver of investor assessment and ultimately entrepreneurial funding. We begin by showing that gender differences in the use of abstraction (Joshi et al., 2020) emerge in the context of pitching entrepreneurial ventures through a study of 129 early-stage entrepreneurs participating in a pitch workshop. Next, we show the implications of these differences for investor perceptions of a startup's long-term growth orientation and its worthiness of investment. To do this, we use both archival data and an experimental study completed by active and experienced investors. In a study of 1,284 investment opportunities presented to a venture capital firm that invests in early-stage startups and in a controlled experiment, we show that divergence in communication style impacts investors' perceptions of long-term growth orientation and influences their willingness to invest in the entrepreneur's venture.

The work we present here seeks to contribute to several areas of research. The first is to the literature on gender and entrepreneurship. We show that founder behaviors, such as the manner in which one communicates, play an important role in entrepreneurial outcomes and may be an important driver of observed gender differences in this arena. Second, we contribute to the work of scholars who have examined entrepreneurial resource acquisition by demonstrating that there are factors that go beyond mere economic criteria in influencing investment decisions—in fact, these factors may indirectly impact the way investors perceive and weigh these economic indicators. Finally, our studies also provide an additional basis for understanding the complex interplay between communication and organizational outcomes and how specific aspects of communication, such as level of abstraction, can have downstream ramifications for individuals and founders. Only with such an understanding can we more fully identify what might be driving gender disparities in entrepreneurial funding and begin to suggest practical ways that entrepreneurs can actively address the disadvantages they may face in seeking resources.

BACKGROUND

Raising funding presents an important challenge for many entrepreneurs, but it seems to be uniquely difficult for female founders. Studies have shown that the "gender penalty" female entrepreneurs face in acquiring financial capital for their ventures is nontrivial (Brooks et al., 2014), with men almost twice as likely as women to receive funding (Webber, 2016). Female-led startups are less likely to be approved for bank loans (Buttner & Rosen, 1988) and are more likely to raise lower amounts of money from investors, such as angel investors and venture capital firms (Kanze, Huang, Conley, & Higgins, 2018), and to receive less favorable financing terms (Fay & Williams, 1993; Jennings & Brush, 2013; Madill, Riding, & Haines, 2006). Given that financial resources directly relate to a firm's ability to survive and achieve financial success (Aldrich & Ruef, 2006), these constraints to accessing financial capital can limit female-founded firms' long-term growth potential (Alsos, Isaksen, & Ljunggren, 2006).

Scholars have provided several possible underlying explanations for gender disparities in entrepreneurial resource acquisition. One body of research has suggested that female entrepreneurs may simply be more likely to lack the managerial or employment experience required for successful venturing (Boden & Nucci, 2000; DeTienne & Chandler, 2007). Despite the human capital differences that may or may not exist, overwhelmingly, scholars have found that *perceived* differences in skills and human capital play a far more prominent role in investor judgments (Huang, 2018; Huang & Pearce, 2015). For example, female entrepreneurs are penalized relative to their male counterparts as a result of role incongruity or a perceived "lack of fit" between female stereotypes and the personal qualities expected of business entrepreneurs (Lee & Huang, 2018), and they are deemed to lack the ability to develop a founding

strategy (Carter, Williams, & Reynolds, 1997) or even assumed to be less focused on achievements and advancement in lieu of risks and prevention (Kanze et al., 2018). Due to these investor biases, female entrepreneurs are often seen as less experienced and less credible in the eyes of investors, and in turn, they are considered less capable of returning the types of financial profits that investors hope for and expect in return for their infusion of capital (Lee & Huang, 2018).

A separate body of literature has argued that regardless of any explicit (or implicit) bias on the part of investors, funding disparities also overwhelmingly exist because of the strategic choices that female entrepreneurs make as compared with male entrepreneurs (Carter, Brush, Greene, Gatewood, & Hart, 2003). Female founders are more likely to start businesses that require less capital, and accordingly, they seek and receive less capital (Becker-Blease & Sohl, 2007; Orser, Riding, & Manley, 2006). Consequently, disparities in funding might also be market driven and based on venture type. Therefore, consistent with previous research, we offer our initial baseline hypothesis:

Hypothesis 1. There is a negative relationship between female entrepreneurs and likelihood of investment, such that female-led ventures are less likely to receive financial capital for their ventures as compared with their male counterparts.

We extend this research by investigating a new mechanism that potentially explains this negative outcome and helps situate prior explanations: communicative style—specifically, an entrepreneur's use of concrete versus abstract language. How an entrepreneur portrays their venture likely plays an important role—alongside presumed investor bias and venture type—in understanding investor judgments of investment worthiness.

THEORY DEVELOPMENT

Research on construal level theory suggests that people can mentally represent information using more abstract or more concrete cognitive representations (for reviews, see Liberman & Trope, 2008; Trope & Liberman, 2010; Wiesenfeld, Reyt, Brockner, & Trope, 2017). Abstract representations (or high-level construals) are relatively decontextualized, focusing on the end goals of an action, an item's or event's overall gist, or macrolevel defining aspects. Concrete representations (low-level construals) are more contextualized and specific, drawing on the means of attaining action and focusing more on incidental features. An individual's communications can also be characterized in terms of their relative level of abstraction, i.e., whether they use more abstract language that captures the bigger picture or more concrete language that captures more specific details (Joshi & Wakslak, 2014; Joshi, Wakslak, Raj, & Trope, 2016). Reliance on abstract versus concrete speech often depends on situational factors. For example, speakers' perceptions of an audience's distance influence speech abstraction, with speakers more likely to use abstract communication when faced with a large and heterogenous audience (Joshi & Wakslak, 2014) or a physically distant audience (Joshi et al., 2016). Furthermore, a speaker's own power and status in a situation also influences their communication style, such that having power is associated with the use of abstract speech and lacking power with the use of concrete speech (Magee, Milliken, & Lurie, 2010; Smith & Trope, 2006).

Building on this body of work, recent research has argued that women are generally more likely than men to speak concretely, and several identified mechanisms converge on this prediction (Joshi et al., 2020). For example, women have historically occupied lower power and status roles compared to men (Eagly & Wood, 1999; Ridgeway, 2001), and the experience of low power has been associated with the use of concrete speech (Magee et al., 2010). Gender differences in the tendency to seek closeness in groups versus establish distance and uniqueness from others (Baumeister & Sommer, 1997) may also play a role in gender differences in speech abstraction, as research has suggested that the motive to establish proximity in communication contexts leads to the use of more concrete communication (Torrez, Wakslak, & Amit, 2019). Further, women's preference for engagement in small and close-knit groups (Fabes, Martin, & Hanish, 2003; Heilman, 1983) may lead women to use more concrete communication because concrete communication is better suited to meet the needs of smaller and more proximal audiences. On the other hand, men's tendency to relate in larger, hierarchical, and more diverse groups (Archer, 2009; Fabes et al., 2003) leads them to use more abstract communication because abstract messages serve to traverse distance and are relatable and relevant to a wider audience (Joshi & Wakslak, 2014; Joshi et al., 2016). Finally, women may be more compelled to demonstrate competence due to stereotypes of incompetence (Fiske, Cuddy, Glick, & Xu, 2002) or due to women's own tendency to internalize these stereotypes and doubt their own competence (Hyde & Durik, 2005; Kumar & Jagacinski, 2006), leading

them to emphasize specifics and details in an attempt to show mastery.

In line with these converging explanations, research has suggested that men and women reliably differ in their use of concrete or abstract speech, an effect that persists across written and verbal communication and higher- and lower-stakes contexts: female participants in a variety of lab studies speak more concretely than male participants, female bloggers speak more concretely than male bloggers, and female members of the U.S. Congress speak more concretely than male members of Congress (Joshi et al., 2020). Of note, however, is that these gender differences in communication do not necessarily reflect ability differences. In fact, when women are primed to think of their audience as being psychologically distant and different from them, the gender difference in abstraction is eliminated, with women and men being equally likely to adapt their speech to the needs of their audience (Joshi et al., 2020).

Given that many of these same aforementioned dynamics are prominent in the entrepreneurship context, it seems likely that a gender difference in the preference for concrete versus abstract speech persists in this context, such that female entrepreneurs would use more concrete speech than male entrepreneurs when describing their venture. First, female entrepreneurs are particularly aware of power and status dynamics in the early-stage investment ecosystem (Bullough, Renko, & Abdelzaher, 2014; Lewis, 2006). Investors, who hold the financial capital that entrepreneurs so desperately need to build and sustain their ventures, wield nearly all the power. As entrepreneurs enter into funding opportunities, statistics showing how rare it is for entrepreneurs to receive funding in general, and for female entrepreneurs in particular, are likely to be top of mind. For example, only about 3% of businesses started each year receive external financing from early-stage investors, and of that, only 2.2% of the 3% goes to female entrepreneurs (Pitchbook & NVCA, 2016). With this acute awareness of the uphill battle that they face, female entrepreneurs often feel that they have minimal power, an experience that leads to the use of more concrete and detail-oriented speech (Magee et al., 2010; Smith & Trope, 2006).

Second, and relatedly, female entrepreneurs feel a larger burden to establish credibility and entrepreneurial belongingness, i.e., to demonstrate that they "fit" the mold of a successful entrepreneur (Lee & Huang, 2018). This may accentuate the tendency to communicate concretely, as female entrepreneurs especially will likely try to emphasize the specific milestones they have accomplished, the discrete actions they have taken, and the details of what they have achieved—each of which is best communicated using a more concrete style (Semin & Fiedler, 1988). Furthermore, in an effort to show that they can continue delivering and attaining subsequent milestones, they likely emphasize *how* they did so in the past. *How* explanations, rather than *why* explanations, are most often indicated through concrete speech (Vallacher & Wegner, 1989).

Third, gender dynamics in the entrepreneurial investment ecosystem often result in female entrepreneurs feeling that they represent members of the out-group and male entrepreneurs feeling like members of the in-group (Bullough et al., 2014; Lewis, 2006). In an effort to establish more of a connection with their audience and circumvent being considered an outsider from the early-stage investors from whom they seek financial capital, female founders will try to form a relational closeness or interdependence with these investors. Attempts to diminish distance and gain closer social proximity are associated with the use of more concrete communication styles (Torrez et al., 2019), and female entrepreneurs may likely further employ this style in investor interactions.

Taken together, the general tendency for women to speak concretely, along with the alignment of female entrepreneurs' (more than male entrepreneurs') interpersonal experiences in entrepreneurship with factors associated with the use of more concrete speech, lead us to suggest:

> *Hypothesis 2. Female entrepreneurs are more likely to use concrete speech (less likely to use abstract speech) than male entrepreneurs.*

Communicative Abstraction and Long-Term Growth Orientation

This tendency toward concrete versus abstract speech likely impacts the perceptions and cognitions of investors who are trying to gauge investment worthiness. According to construal level theory, there is a fundamental link between communicative abstractness and perceptions of "nearness" or "farness" (e.g., distance [Trope & Liberman, 2010; Trope, Liberman, & Wakslak, 2007]). For example, people assume that abstractly described events (e.g., "an academic gathering of management scholars") take place farther into the future than concretely described events (e.g., "we will have scholars presenting academic papers") (Liberman, Trope, McCrea, & Sherman, 2007). More

generally, whereas concrete language relates to observable aspects of items, actions, and events as they exist in the here and now, abstract language instead concerns itself with the meaning of objects and events or the larger purpose and end goals of actions, which has more relevance across variation in time and space (Joshi et al., 2016). This is why speakers use abstract speech when communicating with large or distant audiences, for example—doing so enables them to appropriately connect with people at a greater distance whose concrete experiences might differ from their own (Joshi et al., 2016). In line with this, prior research has suggested that abstract speech, which emphasizes implications that go beyond the current time and place, is associated with perceptions of the speaker as a bigger-picture thinker (Wakslak, Smith, & Han, 2014) and someone more likely to be "visionary" (Carton, 2018).

Based on these prior findings, we posit that an entrepreneur's use of abstract language will be interpreted by investors as an indication that the entrepreneur has a long-term orientation in which he or she is not only focused on the venture as it exists right now but as it might exist in the future. In other words, we argue that because abstract communication emphasizes aspects of items and events that go beyond what is observable in the here and now, entrepreneurs who communicate using a more abstract style will be perceived as forward-looking thinkers who have a broad vision for their venture. Thus, entrepreneurs who use a more abstract style to describe their ventures will be more likely to be perceived by investors as founding startups that are attuned to the long term and oriented toward a broad (rather than localized) strategy.

Such long-term growth-oriented firms—startups that seek to scale and maintain rapid growth over time (Josefy, Kuban, Ireland, & Hitt, 2015)—are distinguished from small businesses and low-growth firms in both their strategy to maximize profits through a long-term orientation on growth and scalability (Gundry & Welsch, 2001), and their reliance on venture funding to help them achieve this growth. Indeed, investors specifically seek out signals that can lead them to conclude that a firm fits into this category (Huang & Knight, 2015; Jin, Wu, & Hitt, 2017; Maxwell, Jeffrey, & Lévesque, 2011). We argue that an entrepreneur's communicative abstraction allows investors to do so by signaling attributes of long-term thinking.

For example, early investors in Uber, a company founded in 2009, described how the founder positioned the company as "a transportation solution that is convenient, reliable, and readily accessible to everyone," rather than "a smartphone app which connects willing passengers to taxi cab drivers" (Isaac, 2019: 172). The former description is highly abstract, while the latter is highly concrete. This use of an abstract style in describing the company (which suggests a solution that is applicable to a widespread audience and one that is not limited in scope) helped position the company as one that was highly oriented toward, and focused on, long-term growth.

Based on the above, we propose that those who speak with a more concrete communication style are more likely to be interpreted as focusing on immediate problems and goals (which are inherent to small businesses and low-growth firms). In contrast, those who speak with a more abstract communication style are more likely to be linked with the type of distal goals that go beyond the present moment, which are typical of firms with a long-term growth orientation. The use of communicative abstraction will be interpreted by investors as an indication that the entrepreneur is focused on this orientation toward the long term. Therefore, we posit:

Hypothesis 3. Entrepreneurs communicating with abstract (concrete) speech are more (less) likely to be perceived as having ventures oriented toward long-term growth.

Long-Term Growth Orientation and Likelihood of Funding

When investors make their investment decisions, among the most important markers they screen for is an entrepreneur's ability to nurture a business opportunity into a potentially high-growth venture; this is an especially critical factor that determines whether resource providers will provide financial capital (Jarillo, 1989; Steffens, Davidsson, & Fitzsimmons, 2009). Investors recognize that not all founders will have a long-term growth orientation. Yet, investors specifically look for those founders who do have this orientation. Indeed, for investors to make the largest possible return on their investment, they depend upon high growth (Mueller, Volery, & Von Siemens, 2012). It is only with a focus on the long term (Busenitz, Fiet, & Moesel, 2005) that ventures have the potential for the type of large-scale growth that produces extraordinary home-run returns (Eisenmann, 2006; Gompers, 1994; Quindlen, 2000), which are the sort of outsized returns that investors need to compensate for the large portion of complete losses in their portfolio (Huang & Pearce, 2015).

Put another way, investors who provide capital to entrepreneurs who do not have a long-term growth orientation are constrained in achieving the types of monetary returns they need. Hence, investors seek to avoid this scenario; entrepreneurs who demonstrate a long-term orientation and the propensity for large-scale growth will be less likely to be screened out of the investment process. Accordingly, we suggest:

Hypothesis 4. Growth orientation is positively related to the likelihood of funding, such that entrepreneurs perceived to have a predisposition for growth are more likely to receive funding for their ventures.

Taken together, investors look for a strong emphasis on long-term firm growth (Steffens et al., 2009) and rely on entrepreneurs to provide them with evidence and signals of this long-term potential. One important signal of long-term growth is communicative abstraction. This may help explain one reason for gender differences in resource acquisition success. If, as we argue, women pitch their ideas less abstractly than men, then they are likely to perceived as being less oriented toward long-term growth. This likely leads to lower amounts of interest from investors interested in such growth-oriented firms. Thus, gender differences in communication style, and in turn, the perception that this communication style elicits from investors about long-term growth orientation, may help explain investors' lower propensity to invest in female entrepreneurs. Therefore, we suggest:

Hypothesis 5. Speech abstraction and perceptions of long-term growth orientation mediate the negative relationship between female entrepreneurs and the likelihood of investment.

OVERVIEW OF STUDIES

We began our investigation with Study 1, which was an initial motivating study in the field to examine whether the general tendency identified in prior work for men to communicate more abstractly than women remains directly applicable in the entrepreneurship context, given the particular considerations, concerns, and constraints that entrepreneurs face. Establishing that this communicative difference in abstraction is indeed observed within the entrepreneurial context provided us with the basis for Studies 2 and 3.

Study 2 then examined a corpus of 1,284 investment opportunities presented to a venture capital firm that invests in early-stage startups to examine the extent to which men opt for more abstract communication and women more concrete communication and the consequences of any such divergence. We tested how communicative abstractness influences whether an investor perceives a venture as oriented toward long-term growth, and in turn, whether an investor is likely to invest. Then, Study 3 took an experimental approach to vary the communicative abstraction of a short venture description that investors evaluate for potential investment. Study 3 helped to rule out the possibility that unobserved differences in ventures pitched by men versus women influence the effect of communicative abstraction on entrepreneurial outcomes.

STUDY 1: GENDERED CONSTRUAL DIFFERENCES IN ENTREPRENEURSHIP PITCHES

In this initial study, we examined whether the general tendency identified in prior work for men to speak more abstractly than women (Joshi et al., 2020) persists in the entrepreneurship context, given the particular considerations, concerns, and constraints that entrepreneurs face. As we explained above, we suspected that this would be the case given the relevance of several factors presumed to underlie the gender-abstraction effect in this context. At the same time, it is plausible that in an entrepreneurship setting, where entrepreneurs are aware that they must communicate the overall vision of their startup, regardless of gender, they may actively strive to use higher levels of abstraction, which would obscure gender differences in communicative abstraction. Therefore, before examining the potential role that the level of abstraction plays on a key investor decision-making factor—the assessment of a venture's orientation toward long-term growth and scalability—we first examined if there were gender differences in the level of abstractness of the language that male and female entrepreneurs use when describing their ventures.

Sample and Procedure

We analyzed a sample of pitches presented by 129 early-stage entrepreneurs (Male = 93, Female = 36)[1]

[1] While only 28% of the entrepreneurs in our sample were women, this proportion far exceeds the typical number of women who participate in pitch competitions and similar funding events, with the percentage of women in these events found to be approximately 9%. Percentages in similar samples ranged from a low of 4% to a high of 23% (Garber, 2013; Huang, Wu, Lee, Bao, Hudson, & Bolle, 2017; Price, 2015).

who participated in a technology entrepreneurship pitch workshop hosted by a nonprofit organization that included a consortium of accredited private investors and angel investor groups in the Mid-Atlantic region in its membership base. A panel of active, experienced angel investors and venture capitalists (VCs) provided suggestions and feedback to each participant at the conclusion of each pitch. In this way, we targeted an event where there were no clear outcomes (i.e., there were no "winners") so that we could first rigorously examine general gender tendencies in how entrepreneurs communicate their ventures.

This setting provided us with a context in which all participants were aware of the salience of long-term scalability as an important factor in the success of their company and provided a consistent sample of entrepreneurs in terms of development stage and industry. All participants were full-time entrepreneurs who founded a technology venture (typically a male-dominated industry). The angel investors and VCs had extensive experience in the technology industry. The university hosting the event also focused on engineering and applied sciences. The entrepreneurs near universally offered technology-based products and services (e.g., information technology, medical devices, or clean energy) and were all at a similarly early stage of development (i.e., prototype development or product market testing).

We coded for speech abstraction in the pitches delivered to the investor panel. Prior to the event, each participant received guidelines to limit their pitch to five minutes and to include six key topic areas: opportunity, customer pain point, market and competitors, financial projections, key risks and solutions, and potential exit options. In this way, the entrepreneurs pitched using the same template but individually chose how to communicate within these bounds. We obtained high-resolution videos of each of the entrepreneurial presentations from this event and subsequently transcribed the videos into text.

Measures: Speech Abstraction

We measured speech abstraction using two different methods to verify the consistency of the results of our prestudy across alternate methods of coding.

Brysbaert Concreteness Index. We measured abstraction using the Brysbaert Concreteness Index (BCI). The BCI relies on abstraction norms that were created for 40,000 commonly used word lemmas in contemporary English (Brysbaert, Warriner, & Kuperman, 2014). Prior research on construal level theory has used the BCI to index speech abstraction of both shorter (Snefjella & Kuperman, 2015) and longer (Bhatia & Walasek, 2016) texts. These concreteness norms emerged from data collected on 4,000 participants who rated the concreteness of different words on a scale ranging from 1 (abstract or language based) to 5 (concrete or experience based). Brysbaert et al. (2014: 906) defined concrete words as referring to "something that exists in reality, you can have immediate experience of it through your senses (smelling, tasting, touching, hearing seeing) and the actions you do." In contrast to concrete words, abstract words are those that are more difficult to visualize and cannot be experienced physically, e.g., *justice*, *morality*, or *strategic vision*. We computed a BCI score for each pitch based on words in the Brysbaert dictionary, calculating both a mean estimate of the overall concreteness of the language used in the pitch as well as an estimate of the overall variability in linguistic concreteness in the particular pitch.[2] Examples of pitches that were rated as abstract or concrete are shown in Appendix A.

Linguistic category model. We also measured abstraction using a second method, the linguistic category model (LCM) (Coenen, Hedebouw, & Semin, 2006; Semin & Fiedler, 1988). The LCM assumes that different linguistic categories vary in their level of abstraction. Prior construal level theory research used LCM to code language for abstraction level (Fujita, Henderson, Eng, Trope, & Liberman, 2006; Joshi & Wakslak, 2014). We used this second coding method to confirm the robustness of our findings across different implementations of linguistic abstraction. A manual coder, blind to the study variables, identified words in four linguistic categories falling along an abstractness continuum, from least abstract (most concrete) to most abstract (least concrete): (a) descriptive action verbs, e.g., *go* or *walk*; (b) interpretive action verbs, e.g., *help* or *socialize*; (c) state verbs, e.g., *love* or *admire*; and (d) adjectives, e.g., *busy* or *great*. At the most concrete end of the spectrum, descriptive action verbs describe an observable action with a clear beginning and end. More

[2] We used the Recursive Inspection of Text (RIOT) scan to generate the Brysbaert Concreteness Index (Boyd, 2016). The RIOT scan is a freely available software tool for calculating indices of language from text files. More information on the RIOT scan can be obtained here: https://riot.ryanb.cc/. Like other researchers using this approach, the RIOT scan methodology does not account for bigrams in the Brysbaert dictionary (Bhatia & Walasek, 2016; Snefjella & Kuperman, 2015).

abstract than descriptive action verbs, interpretive action verbs have a beginning and an end but require interpretation; they thus tie less directly into a single referent activity. Yet more abstract, state verbs, including state action verbs, describe an enduring emotional or mental state rather than a physical action. Finally, at the most abstract end, adjectives are the most decontextualized words and are disconnected from a single action referent. In line with recommendations provided in the LCM manual, we constructed an overall abstraction score ranging from 1 to 4 for each participant using the data from these linguistic categories in the following weighted index:

$$\text{Abstraction Score} = \frac{4\times\text{Adj}+3\times\text{State Verb}+2\times\text{Interp. Verb}+\text{Desc. Verb}}{\text{Adj}+\text{State Verb}+\text{Interp. Verb}+\text{Desc. Verb}}$$

Robustness checks. To assess the robustness of our findings relating gender to differences in speech abstraction, we also ran tests relating gender to other speech characteristics that may be associated with speech abstraction. We constructed the following measures from the pitches: speech tentativeness, e.g., *perhaps*, *guess*, or *maybe* (Linguistic Inquiry and Word Count [LIWC] dictionary [Pennebaker, Booth, & Francis, 2007]); certainty, e.g., *always* or *never* (LIWC dictionary [Pennebaker et al., 2007]); power, e.g., *impunity*, *dominate*, *expertise*, or *status* (Harvard General Inquirer [Stone, Bales, Namenwirth, & Ogilvie, 2018]); affiliation, e.g., *care*, *trust*, or *child* (Harvard General Inquirer [Stone et al., 2018]); number of sentences; and total word count.

Results

Across both the BCI method and the LCM method, we found that men used more abstract language in their pitches than women, providing evidence that gender differences in communication also exist in the entrepreneurship context (Table 1). Using the BCI method, an independent samples *t*-test computed on the BCI scores[3] showed that men (M = 2.47, SD = .08) pitched their ideas less concretely than women (M = 2.51, SD = .10), $t(128) = 2.15$, $p < .05$, $d = .38$. Using the LCM, an independent samples *t*-test on the abstraction score also confirmed that men communicated their entrepreneurial ideas more abstractly (M = 2.66, SD = .17) than women (M = 2.56, SD = .16), $t(128) = 3.10$, $p < .01$, $d = .60$.

Thus, we observed differences in the degree to which men and women used linguistic abstraction in the entrepreneurship context, supporting Hypothesis 2. In addition, women (M = 1.00, SD = .07) exhibited more variability in their speech in terms of abstraction than men (M = .97, SD = .07), $t(128) = 2.02$, $p < .05$, as measured by intraindividual variability in the BCI. We found no significant gender differences in the use of tentative language, certain language, power words, or affiliation words; we also found no statistically significant gender differences in speech length as measured by word count or number of sentences spoken (see Table 1).

Discussion

We examined whether the gender differences in communicative style found in prior research would hold in the entrepreneurship context and found that female entrepreneurs described their ventures more concretely than male entrepreneurs. Establishing that this communicative difference in abstraction observed in prior research holds within entrepreneurship provides us with the basis to examine the consequences of these gender differences in communication style for entrepreneurial outcomes.

TABLE 1
Study 1: Gender Differences in Language Use

	Men		Women		
	Mean	*SD*	Mean	*SD*	*t*
Word count	428.87	(159.06)	383.22	(135.30)	1.52
Sentences	26.43	(11.21)	23.83	(7.29)	1.29
Brysbaert Concreteness Index	2.47	(0.08)	2.51	(0.10)	2.15*
Brysbaert Variance Index	0.97	(0.07)	1.00	(0.07)	2.02*
Linguistic abstraction (LCM)	2.66	(0.17)	2.56	(0.16)	3.10**
Tentativeness words	1.89	(1.04)	2.03	(1.53)	0.62
Certainty words	1.07	(0.61)	0.91	(0.51)	1.43
Power words	3.44	(1.24)	3.64	(1.56)	0.73
Affiliation words	4.67	(1.90)	4.61	(2.16)	0.17
Observations	93		36		

Note: LCM = linguistic category model.
* $p < .05$
** $p < .01$

[3] We note that the concreteness norm for each of the 40,000 words that encompass the Brysbaert Concreteness Index is derived from a rating on a five-point scale. Hence, the scores that are based on words used in naturally produced language will tend to cluster together to the scale mean of 2.5, and small deviations from these means are in fact meaningfully different. See Snefjella and Kuperman (2015) to confirm the relatively small variance of speech concreteness scores in naturally occurring speech.

STUDY 2: FIELD STUDY OF A VC INVESTMENT FIRM

Sample and Procedure

We studied 1,284 funding applications submitted by startup companies to a prominent VC firm based on the West Coast of the United States. This sample of applications from 243 female founders and 1,041 male founders represented the entire corpus of all funding requests received by this VC firm in 2018.

As part of the submission process, the startup companies submitted their application through an online submission platform. In addition to the application itself, included on the submission platform was information on the VC firm's investment philosophy, specific investment criteria, and FAQs. This information explicitly stated that the VC firm considered startups that fulfilled the following criteria: "up to a $2M raise with $5M–$10M total to potential exit"; "a pre-money valuation less than $5M, with higher pre-money reducing probability of funding"; and "equity strongly preferred, rather than convertible notes." These parameters indicated that for startups deemed worthy of investment, the VC firm would invest up to $2 million in funding in the current period and would consider reinvesting additional capital in subsequent rounds of funding up to $5–$10 million. Further, the VC firm stated a specific interest in companies considered to be worth less than $5 million based on their current revenues and their current offerings. Finally, the VC firm expressed an explicit interest in making an equity investment—which is a form of capital investment that exchanges cash for an ownership stake in the company—rather than a debt investment such as a convertible note.

Each of these qualifications was important because they provided consistency in the type of company the VC firm sought to fund: early-stage companies interested in long-term growth and scalability rather than those with a more short-term orientation. Furthermore, the VC firm specified that they were looking for "underserved, disruptive market potential with large upside," which is additional evidence of long-term growth and scale as the VC firm's investment priorities. In prior funds, this firm typically invested in four to six new deals with 10–12 follow-on investments in portfolio companies.

To be considered for funding, applicants first submitted an initial application, which required them to complete basic information about the startup (location, team members, and industry focus), as well as an executive summary of the company. Founders were told that once these materials were submitted, the VC firm would perform an initial screening to ensure that the company fell within their scope of interest, and that if they made it past this initial cut, they would be contacted for additional information and further due diligence. Specifically, they were told: "Should your company pass our review stage, we will reach out to you and conduct a series of due diligence exchanges with our screening committee." To test the hypotheses, we coded the communicative abstraction of each executive summary submitted in this initial stage.

Measures

Gender. The application included a measure of founder gender, which we used as a dummy variable indicating whether the entrepreneur was female (Female = 1, Male = 0). The VC firm specified that the application should be completed by the founder or CEO, who would also be the primary contact.

Level of abstraction. We computed an overall concreteness score for each entrepreneur's written proposal using the BCI, following the same procedure detailed in Study 1. We calculated both a mean estimate of the overall concreteness of the language used in each application as well as an estimate of the overall variability in linguistic concreteness to capture individual-level variability in the particular application. We then reverse-scored the BCI concreteness measure to provide a measure of abstraction and facilitate the interpretation of any effect.

Long-term growth orientation. Perceptions of long-term growth and scalability were measured using a three-item scale adapted from validated scales of entrepreneurial growth and scalability (Cardon, Mitteness, & Sudek, 2017; Walter, Auer, & Ritter, 2006). On a Likert-type scale that ranged from 1 (not at all) to 5 (a lot), investors rated each startup on the following items: "large growth potential," "long-term revenue potential," and "highly scalable." Cronbach's α was 0.89.

Likelihood of investment. Investors performed an initial screening, upon which approximately 90% of startups were eliminated from consideration. We assessed investment likelihood based on a binary measure of whether the startup was one of the approximately 10% of startups (122 out of the total 1,284 startups) that made it through the initial stage of funding consideration.

Control variables. We gathered variables related to the nature of the venture that may be associated with differences in how entrepreneurs pitch, including the industry focus, whether the venture was

TABLE 2
Study 2: Means, Standard Deviations, and Correlations

	Variable	Mean	*SD*	1	2	3	4	5	6	7
1	Gender (female = 1, male = 0)	0.19	0.39							
2	Team size	2.17	0.91	0.02						
3	Industry	3.51	1.71	−0.01	−0.02					
4	Business target (B2B = 1, B2C = 2)	1.56	0.50	−0.02	0.01	0.01				
5	Business offering (service = 1, product = 2)	1.50	0.50	−0.01	0.03	−0.06*	−0.02			
6	Communicative abstraction	2.56	0.18	−0.19	0.01	−0.09**	0.01	0.19**		
7	Long-term growth orientation	2.98	0.80	−0.10	−0.01	0.09**	0.01	−0.11**	0.49**	
8	Investment likelihood	0.10	0.29	−0.06	0.02	0.03	−0.01	−0.06*	0.09**	0.08**

* $p < .05$
** $p < .01$

targeting a consumer (i.e., business-to-consumer [B2C]) versus business (i.e., business-to-business [B2B]) market, and whether the main offering was a product or service. We also controlled for team size as a venture-specific factor that may affect how entrepreneurs choose to pitch their venture.

Analyses and Results

Descriptive statistics and correlations between variables are reported in Table 2. Table 3 reports the regression analyses. For models with the dichotomous dependent variable measuring investment likelihood, we use a binary logistic regression model and report a regular nonstandardized coefficient estimate (*b*). For the other models with continuous dependent variables measuring communicative abstraction and long-term growth, we use an ordinary least squares regression model and report standardized coefficient estimates (β).[4] For simple slope analyses reported in the text and the mediation analysis shown in Figure 1, we report regular nonstandardized estimates.

We used binary logistic regression to test our baseline Hypothesis 1, which posited that female entrepreneurs were more likely to be eliminated from funding consideration after the initial investor screening. As seen in Table 3, results show a negative and significant coefficient of female gender of the entrepreneur in Model 1 ($b = -.63$, odds ratio = .53), which suggests that the odds of female-led ventures making it through the initial stage of consideration for funding was only 53% of that for male-led ventures.

We employed linear regression to test Hypotheses 2 and 3. As shown in Table 3 (Model 2), our prediction (Hypothesis 2) that female entrepreneurs would present their venture less abstractly (more concretely) was supported ($\beta = -.19$, $t = -7.06$, $p < .001$). Repeating the procedure from Study 1, we also tested for other gendered communication patterns that prior research has suggested might impact entrepreneurs seeking access to entrepreneurial funds (e.g., Lee & Huang, 2018). We found no significant gender differences in the tendency to use tentative language, certain language, power words, or affiliation words. We also found no significant evidence that male and female entrepreneurs differed in word count or speech length or on other established dimensions along which perceptions of men's and women's language tend to differ (e.g., intimacy versus dominance and warmth versus agency).

Hypothesis 3 stated that abstract speech would positively impact perceptions of growth orientation, and the results revealed that this was the case ($\beta = .48$, $t = 19.29$, $p < .001$; Model 3). We also found support for Hypothesis 4, whereby the results of a binary logistic regression revealed that perceptions of growth orientation were positively associated with investment likelihood (i.e., making it through the initial stage of consideration) ($b = .37$, odds ratio = 1.45; Model 4). We note that we found support for Hypothesis 4, in addition to each of our hypotheses, when our control variables were omitted.

We next tested Hypothesis 5, which posited that speech abstraction and growth potential mediate the relationship between gender and the likelihood of funding. To calculate the simple indirect effect, we constructed a 95% bias-corrected bootstrap confidence interval, and we found that the indirect effect of gender on investment likelihood—through communication abstraction and in turn long-term orientation—was significant (indirect effect = −.03, $SE = .02$, CI_{95} [−.08, −.01]).

After examining this indirect effect occurring via communicative abstraction and perceptions of long-

[4] Because of the standardized coefficients, these models do not generate a constant term coefficient to report.

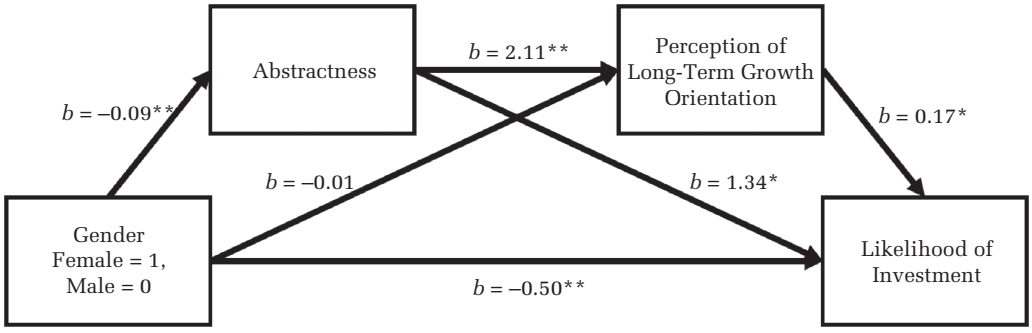
TABLE 3
Study 2: Results

Variables	Model (1)		Model (2)		Model (3)		Model (4)		Model (5)		Model (6)	
	Investment Likelihood		Communicative Abstraction		LT Growth Orientation		Investment Likelihood		LT Growth Orientation		Investment Likelihood	
	b	***SE***	β	***t***	β	***t***	***b***	***SE***	β	***t***	***b***	***SE***
Constant	−2.10***	0.53					0.73	1.97			0.70	1.97
Team size	0.05	0.11	−0.01	−0.06	−0.01	−0.04	0.06	0.11	−0.01	−0.04	0.70	0.06
Industry	0.03	0.06	0.09***	3.30	0.05*	2.02	0.01	0.06	0.05*	2.01	0.06	0.06
Business target	−0.03	0.19	−0.01	−0.52	0.01	0.50	−0.02	0.19	0.01	0.50	−0.03	0.19
Business offering	−0.15	0.19	−0.01	−0.47	0.04*	1.75	−0.16	0.19	0.04	1.72	−0.16	0.19
Gender (female = 1, male = 0)	−0.63*	0.29	−0.19***	−7.06	−0.01	−0.25	−0.51*	0.30	−0.23	−0.71	0.01	1.19
Communicative abstraction					0.48***	19.29	1.31*	0.67	0.49***	17.10	−1.33*	0.07
LT growth orientation							0.37*	0.14			0.19	0.14
Gender × communicative abstraction									−0.23	−0.70		
Gender × LT growth orientation											0.13*	0.05
Log likelihood	799.81						789.35				789.17	
χ^2	5.54						16.98*				17.17*	
F			12.34***		67.53***				57.93***			
R^2			0.05		0.24				0.24			
df	5		5		6		7		7		8	
Observations	1,284		1,284		1,284		1,284		1,284		1,284	

Notes: Models (1), (4), and (6) use a binary logistic regression and report nonstandardized estimates (*b*) and standard errors (*SE*). Models (2), (3), and (5) use an ordinary least squares regression and report standardized estimates (β) and *t*-statistics (*t*). Tests are one-tailed.

* $p < .05$
** $p < .01$
*** $p < .001$

FIGURE 1
Study 2: Illustration of Serial Mediation Effect



Indirect Effect	
Abstractness	–0.12*
Perception of Long-Term Growth Orientation	0.01
Likelihood of Investment	–0.03*

* $p < .05$
** $p < .01$

term growth, we found that there remained a significant direct effect of gender on investment likelihood ($b = -.51$; $p = .05$). This suggested that our proposed indirect effect (Hypothesis 5) did not fully account for the effect of gender on investment likelihood (i.e., partial mediation) and that there may be additional indirect effects.

As such, we then conducted an additional supplemental test to understand more fully the multiple indirect effects that might exist (Figure 1). To test for the multiple-step mediating effects of communicative abstraction and perceptions of long-term growth orientation, we utilized the PROCESS multiple mediation model 6 (Hayes, 2012). PROCESS is a conditional process modeling program that utilizes an ordinary least squares- or logistic-based path analytical framework to test for both indirect and direct effects (Hayes, 2012). Figure 1 depicts the two-mediator model in which the predictor variable gender (X) is modeled as affecting the outcome variable likelihood of investment (Y) through four pathways. One indirect pathway runs from X to Y through the first mediator abstractness (M1) only, a second indirect path runs through the second mediator perception of long-term growth (M2) only, and a third indirect pathway passes through both M1 and M2 sequentially, with M1 affecting M2. All indirect effects were subjected to bootstrap analyses with 10,000 bootstrap samples and the calculation of 95% confidence intervals (CIs; see Table 3). A bias-corrected bootstrap 95% CI for the product of these paths that does not include zero provides evidence of a significant indirect effect (Preacher & Hayes, 2008).

Gender exerted an effect on communicative abstraction ($b = -.09$, $t = -7.06$, $p < .01$) and communicative abstraction then impacted perceptions of long-term growth orientation ($b = 2.11$, $t = 19.50$, $p < .01$), which in turn influenced investment likelihood ($b = .17$, $t = 12.45$, $p < .05$). As shown in Figure 1, communicative abstraction and perceptions of long-term growth orientation mediated the negative effect of female entrepreneurs and funding outcomes based on a bias-corrected bootstrapped confidence interval that did not contain zero (indirect effect = $-.03$, $SE = .02$, CI_{95} [$-.08$, $-.01$]), and the indirect path that ran through communicative abstraction also did not contain zero (indirect effect = $-.12$, $SE = .03$, CI_{95} [$-.26$, $-.01$]). However, the indirect path that ran through perceptions of long-term growth only (M2) did contain zero (indirect effect = $-.01$, $SE = .01$, CI_{95} [$-.03$, .01]).

We also conducted a post hoc exploratory test to examine how gender might moderate the relationships between communicative abstraction, perceptions of long-term growth orientation, and investment likelihood. That is, rather than treating gender as the first causal link in a chain, as we did above, we considered whether gender might influence the relationships between the other variables. We did not have any a priori hypotheses for these effects and considered them in an exploratory manner. We first examined whether gender moderates the effect of communicative abstraction on perceptions of

long-term growth orientation and did not find a significant interaction between gender and abstraction (see Table 3, Model 5). We next examined whether gender moderates the effect of perceptions of long-term growth orientation on investment likelihood and found that there was a significant interaction ($b = .13$, $p < .05$; see Table 3, Model 6). Simple slope analyses (Aiken & West, 1991) revealed that perceptions of long-term growth were positively related to investment likelihood for female-led ventures ($b = .05$, $p < .05$) and male-led ventures ($b = .18$, $p < .05$), but the significant interaction indicates that the effect was stronger for male-led ventures.

Discussion

Findings from Study 2 replicate and build upon those of Study 1. As in Study 1, we found that female entrepreneurs—here, those applying for funding from a VC company rather than students in a university pitch competition—use more concrete language than their male counterparts. More critically, supporting Hypotheses 3, 4, and 5, we found that this gender difference has important implications: more abstract speech was associated with greater perceived growth potential and a higher likelihood of making it to the next round of consideration, with these variables mediating the gender–investment decision link.

Furthermore, while not explicitly hypothesized, the broader multiple mediation model which considers all possible mediating paths in the sequential mediation suggested two additional things. First, we found that the direct effect of gender on investment outcomes persisted when controlling for the entrepreneur's communicative abstraction and perceptions of long-term growth orientation (i.e., the indirect effects). This is not especially surprising given other documented mechanisms for gender disparity in funding outcomes that we do not empirically consider here (see our earlier discussion in the Background section). Second, we found a second indirect effect of gender on investment via abstraction only. This suggests that something in addition to perceived long-term orientation explains the impact of abstraction on investment outcome. While speculative, a possibility based on the prior literature is that abstraction signals power and leadership capability (Wakslak et al., 2014) as well as expertise (Reyt, Wiesenfeld, & Trope, 2016), both of which may be important attributes of successful founders. Future research can more meaningfully assess the varied paths that might link communicative abstraction and investment decisions in this context.

Also interesting were our exploratory analyses of the moderating role of gender. We did not find that gender moderated the relationship between abstraction and perceptions of long-term orientation. However, gender did moderate the relationship between perceptions of long-term orientation and funding outcomes. While not based on a priori predictions, this suggests that perceptions of long-term growth orientation are a salient consideration in investor decisions to allocate financial capital, but that it may be even more important for female entrepreneurs, relative to their male counterparts, to make additional explicit efforts toward conveying their long-term orientation through other means in order to raise financing.

One major limitation of Study 2 is the correlational nature of the study, which leaves open a number of alternative explanations for our findings. Therefore, the next study seeks to provide a second test of our hypotheses with causal evidence in support of these arguments and also rule out alternative possibilities with an experimental approach in which seasoned investors reacted to an abstract or concrete pitch for the same venture.

STUDY 3: EXPERIMENTAL STUDY

The findings from Study 2 support gender as an antecedent to speech abstraction and demonstrate that the level of abstraction impacts perceptions of long-term growth orientation, and in turn, impacts entrepreneurial investors' investment decisions. However, as mentioned earlier, a key limitation of Study 2 was our use of a correlational design, which leaves our findings open to several alternative explanations. For example, men and women might found and pitch fundamentally different kinds of ventures, which gives rise to differences in communicative abstraction and in turn results in differences in which ventures receive investor funding. Or, certain kinds of ventures may tend to be described by their founders more abstractly, and these same ventures might have higher growth potential. Thus, the primary goal of Study 3 was to provide causal evidence in support of our hypotheses and rule out such alternative possibilities.

We sought to do this by testing the effect of abstraction on perceptions of long-term growth orientation while controlling for differences in types of ventures and pitches. That is, we experimentally manipulated the level of abstraction of a baseline pitch so that we

could isolate the impact of communicative abstraction on perceptions of long-term growth orientation and in turn the likelihood of investment. Because the causal effect of gender on the level of abstraction is more well-established, we focused Study 3 on this latter causal effect of abstraction on investors' perceptions and investment decisions. Furthermore, the focal mediator of interest that explains gender differences in entrepreneurial outcomes that we establish in this paper is communicative abstraction. We sought to examine the causal effect of this variable by experimentally manipulating (Stone-Romero & Rosopa, 2010) communicative abstraction rather than measuring it—an approach we used in the previous study. As we articulated earlier, investors look for a strong emphasis on long-term firm growth (Steffens et al., 2009), and communicative abstraction may provide an important signal of this long-term orientation. Perceptions that this communication style elicits from investors about long-term growth orientation should in turn affect investors' likelihood of making an investment. Therefore, we suggest:

Hypothesis 6. Perceptions of growth-orientation mediate the relationship between speech abstraction and likelihood of investment.

Sample and Participants

We recruited 238 active and experienced investors (143 male, 93 female)[5] to evaluate an abstractly or concretely described venture. The participants, each of whom was an accredited investor, had an average of 15.1 years (SD = 11.5 years) of investment experience and 5.5 previously executed investments (SD = 4.1 investments). In the United States, accredited investors have a net worth of at least $1 million, excluding the value of their primary residence, or have an income of at least $200,000 each year for the last two years. Each investor was an attendee at a large entrepreneurial investment conference held annually in different locations across the United States. The conference encompassed a number of keynote speeches as well as breakout workshops with relevant content on entrepreneurial trends and investment practices. At the conclusion of each workshop, investors were asked to complete a brief survey with their opinions of the workshop. This survey included an additional request asking for their willingness to participate in a short study that investigated investment behaviors. In return for their participation, investors were offered the opportunity to download a separate white paper (i.e., research report) summarizing prior research in this area.

[5] Two investors chose not to indicate their gender.

Design and Procedure

We presented investors with a short description of an entrepreneurial venture and asked them to subsequently evaluate the venture. Investors were told that the venture was an actual startup and that they should put themselves in a state of mind where they imagined actually considering the venture for their own investment portfolio.

We manipulated the style of the pitch so that investors received either a concrete or abstract style of pitch. To do so, information in the description of a baseline venture varied (see Appendix B for the full pitch descriptions). For instance, in the concrete condition, investors were told that "the Nanospray coating produces a water-resistant layer only around 100 nanometers (15–30 molecules) thick that can bond to any surface, including glass, wood, tile, brick, and stone." In the abstract condition, investors were instead told that "the Nanospray coating is highly flexible and breathable and can bond to a wide range of surfaces, protecting it from dirt, bacteria, UV light, and acids." Extensive pretesting of the base stimuli was conducted on a nonoverlapping sample of 150 participants (82 male, 68 female) recruited through Amazon's Mechanical Turk to ensure that the venture industry and the basis for the venture idea were gender neutral and that neither was perceived as male typed or female typed. Furthermore, we conducted additional pretests of each description on a nonoverlapping sample of 190 participants to ensure that we were indeed depicting a concrete versus abstract style of communication. Investors in the concrete condition reported feeling that the pitch focused on the details of the venture (M = 4.13, SD = 1.12) more than investors in the abstract condition (M = 3.11, SD = 1.32; $t(189) = 5.76$, $p < .01$), and investors in the abstract condition reported feeling that the pitch focused on the broader goals of the venture (M = 4.61, SD = 1.28; for the concrete condition, M = 2.42, SD = 1.45; $t(189) = 11.01$, $p < .01$).

In addition to varying the abstractness of the pitch, we also varied the gender of the founder via the founder's first name (Jennifer Martin versus Jonathan Martin), so that the findings are not specific to one gender. Thus, investors effectively received one of four randomly assigned versions of the startup

description: (a) female entrepreneur pitching concretely, (b) male entrepreneur pitching concretely, (c) female entrepreneur pitching abstractly, or (d) male entrepreneur pitching abstractly in a two (concrete versus abstract style of pitch) by two (female versus male entrepreneur) experimental design. Although we do not include hypotheses regarding the moderating effect of gender, this manipulation also allows us to conduct additional exploratory analyses using gender as a moderator. Given that exploratory analyses in Study 2 suggested that gender moderated the relationship between long-term orientation and funding decisions, it is especially useful to be able to conduct similar analyses in the context of an experimental study.

After reading the description of the venture, investors were asked to fill out a questionnaire to indicate their perceptions of the venture. They were also asked to report demographic information about themselves.

Measures

Perceptions of long-term growth. Perceptions of long-term growth and scalability were measured in similar fashion to Study 2 (Cardon et al., 2017; Walter et al., 2006) using a two-item scale that sought to distinctly capture investors' perceptions of long-term growth orientation. Items included the following: "To what extent do you see this as a highly scalable, high-growth company?" and "To what extent do you agree that this company has long-term growth potential?" (α = .86). Investors rated each of these items on a five-point Likert scale.

Likelihood of investment. Investors indicated their likelihood of making an investment in the entrepreneur on a five-point Likert-type scale in response to a single item: "This is a company that you would invest in: 1 = strongly disagree to 5 = strongly agree." We used this straightforward measure of investment propensity following past research that has found that simple measures best capture the inherent variability in early-stage entrepreneurial progress and long-term outcomes (Huang & Pearce, 2015; Kerr, Lerner, & Schoar, 2011; Kim, Longest, & Lippmann, 2015).

Additional controls. We included a number of controls to account for investor-specific differences, including investor gender, number of years of investment experience, and number of prior investments made.

Analyses and Results

Manipulation checks. After completing the main dependent variables, participants were asked to rate the concreteness of the pitch by providing their agreement on a 1 (strongly disagree) to 5 (strongly agree) Likert scale of the following items: "The pitch included many details about the venture" and "The pitch provided specifics about the venture (α = 0.84). An analysis of variance (ANOVA) revealed concreteness to be significantly affected by treatment condition, $F(1,237) = 2.01$, $p < .05$, with pitches in the concreteness condition rated higher ($M = 3.42$) than pitches in the abstractness condition ($M = 3.11$; $p < .05$). We also asked investors to indicate the gender of the entrepreneur (e.g., after the study, investors were asked, "What was the gender of the entrepreneur you just read about?"), and 100% of investors correctly recalled entrepreneur gender.

In addition to these manipulation or attention checks within the focal study, we also conducted an additional post hoc test on a separate sample of 100 employed participants from Amazon's Mechanical Turk (Buhrmester, Kwang, & Gosling, 2011) to verify that there were significant differences in perceptions of concreteness across our concrete ($M = 4.25$, SD = .89) and abstract conditions ($M = 3.53$, $SD = 1.01$); $p < .01$, but no mean differences on other potentially relevant factors such as perceptions of business expertise ($F = .98$, $p = .33$), scientific expertise (F = 1.71, $p = .19$), enthusiasm ($F = 1.41$, $p = .24$), and optimism ($F = 3.81$, $p = .05$).

Hypothesis testing. Descriptive statistics and correlations between variables are reported in Table 4. Regression results appear in Table 5. To test our main hypotheses, we used ordinary least squares regressions and report standardized coefficient estimates (β) in Table 5. Analyzing the data with ANOVA produced no differences in statistical conclusions. For simple slope and mediation analyses reported in the text, we report regular unstandardized estimates (b).

Hypothesis 3 states that entrepreneurs communicating with abstract speech would be more likely to be perceived as having a long-term growth orientation. Communicative abstraction was positively related to perceptions of long-term growth ($\beta = .12$, t = 1.95, $p < .05$; Model 2). To examine Hypothesis 4, we tested the relationship between perceptions of long-term growth orientation and investment likelihood and found a positive relationship ($\beta = .65$, t = 11.76, $p < .001$; Model 4) in full support of our hypothesis. Again, we note that we found support for

TABLE 4
Study 3: Mean, Standard Deviations, and Correlations

	Variable	Mean	*SD*	1	2	3	4	5	6
1	Investor gender (female = 1, male = 2)	1.39	0.49						
2	Investor years of experience	15.00	10.98	0.06					
3	Investor number of investments	5.74	4.41	0.07	0.56**				
4	Entrepreneur gender (female = 1, male = 2)	1.45	0.50	0.03	−0.01	0.01			
5	Communicative abstraction (concrete = 1, abstract = 2)	1.59	0.49	0.02	0.10	−0.03	0.06		
6	Perceptions of long-term growth orientation	3.50	0.85	0.05	0.03	−0.05	0.35**	0.13*	
7	Likelihood of investment	3.27	1.12	0.13*	0.03	−0.05	0.14*	0.11*	0.59**

* $p < .05$
** $p < .01$

TABLE 5
Study 3: Results

Variables	Model (1)		Model (2)		Model (3)		Model (4)		Model (5)	
	Investment Likelihood		Perceptions of LT Growth		Perceptions of LT Growth		Investment Likelihood		Investment Likelihood	
	β	*t*	β	*t*	β	*t*	β	*t*	β	*t*
Investor gender	0.10	1.54	0.03	0.50	0.03	0.46	0.12*	2.33	0.11*	2.02
Investor years of experience	0.07	0.85	0.09	1.26	0.08	1.10	0.01	0.10	−0.02	−0.27
Investor number of investments	−0.07	−0.96	−0.10	−1.36	−0.08	−1.08	−0.01	−0.15	0.00	0.02
Entrepreneur gender (manipulation)	0.12*	1.80	0.35***	5.57	0.79***	4.32	0.11*	1.92	0.55*	2.34
Communicative abstractness (manipulation)	0.11	1.51	0.12*	1.95	0.34*	1.79	0.18***	3.39	0.16**	3.04
Gender × abstraction					0.64**	2.58				
Perceptions of LT growth orientation							0.65***	11.76	0.96***	5.64
Gender × perceptions of LT growth orientation									0.46*	1.94
F	2.16*		7.76***		7.74***		25.91***		23.01***	
df	5		5		6		6		7	
R^2	0.05		0.14		0.17		0.40		0.41	

Notes: All models use an ordinary least squares regression and report standardized estimates (β) and *t*-statistics (*t*). Tests are one-tailed.
* $p < .05$
** $p < .01$
*** $p < .001$

Hypothesis 4, in addition to each of our hypotheses, when our control variables were omitted.

Unlike in Study 2, where abstraction could freely vary, here, we experimentally varied levels of abstraction in the entrepreneurial pitches. Thus, we could not look at serial mediation in this context. Therefore, we next investigated Hypothesis 6—that perceptions of growth orientation mediate the relationship between speech abstraction and the likelihood of investment—by constructing a 95% bias-corrected bootstrap confidence interval (CI_{95}). In support of Hypothesis 6, communicative abstraction had a significant indirect effect on investment likelihood through perceptions of long-term growth orientation ($b = .29$, $SE = .12$, CI_{95} [.16, .36]).

Given that we orthogonally varied the gender and level of abstraction of the pitch, we further conducted exploratory tests to examine again how gender might moderate these effects. We first examined whether gender moderates the effect of abstraction on perceptions of long-term growth orientation and found a significant interaction between gender and abstraction ($\beta = .64$, $p < .01$; Model 3). Simple slope analyses (Aiken & West, 1991) revealed that the level of abstraction was positively related to perceptions of long-term growth orientation for female-led ventures ($b = .31$, $t = 3.44$, $p < .01$, but the slope for male-led ventures was insignificant ($p = .32$). The significant interaction indicates that how men communicated their ventures did not conclusively impact their likelihood of being perceived by investors as oriented toward long-term growth, and the effect was instead driven by the impact of abstract communication influencing perceptions of long-term

growth for female-led ventures. We next examined whether gender moderates the effect of long-term growth on investment likelihood, and consistent with Study 2, we found a significant interaction (β = .46, $p < .05$; Model 5). Simple slope analyses revealed that perceptions of long-term growth were positively related to investment likelihood for female-led ventures ($b = .52$, $t = 5.46$, $p < .01$) and male-led ventures ($b = .77$, $t = 13.22$, $p < .01$), but the significant interaction indicates that the effect was stronger for male-led ventures.

While exploratory, these findings suggest that an entrepreneur's gender might differentially impact investor perceptions, both in terms of how investors consider perceptions of long-term growth orientation as well as how perceptions of long-term growth orientation influence their investment decisions. Speaking abstractly may provide an avenue for women to influence investor perceptions of their long-term growth orientation. Female entrepreneurs absorb a larger penalty, however, relative to men, in terms of how these perceptions of long-term growth orientation impact their propensity to receive investment. We discuss the implications of this finding and the importance of further research in this area in more detail below.

Discussion

In a controlled experiment of investors vetting investment opportunities, we further shed light on the key role of communicative abstraction in determining investor perceptions of long-term growth and scalability. We again found that framing a pitch abstractly led to higher perceptions of long-term growth and scalability, which in turn resulted in a greater willingness to invest in the startup.

In addition, in exploratory analyses, we found that the implications of speaking concretely, rather than abstractly, were more pronounced for female entrepreneurs based on the subsequent perceptions and attributions made by investors, further shedding light on how the VC funding process may not be entirely devoid of gender bias on the part of investors. Specifically, our findings suggest that men seem to receive some benefit of the doubt, whereby they are more likely to be assumed to be creating long-term, high-growth firms, even when they communicate relatively concretely. Perceptions of female founders and their orientation toward long-term growth, on the other hand, are more prevalently driven by the associations that investors make based on communicative abstraction. We discuss the implications of this differential in perceptions of long-term growth in more detail in the General Discussion section.

GENERAL DISCUSSION

Our findings demonstrate that there are gender differences in how men and women pitch ventures, with female entrepreneurs communicating their ideas less abstractly than male entrepreneurs. Furthermore, we showed that investors not only rely on communicative abstraction as an indicator of the long-term growth and scalability of a venture opportunity, but they are also more willing to invest in ideas that are framed abstractly rather than concretely. Thus, women's tendency to speak less abstractly than men helps partially explain the gender disparity that exists in entrepreneurial investment.

Theoretical Implications

Our findings contribute to several lines of theoretical inquiry including research on gender disparities in entrepreneurship, entrepreneurial resource acquisition, and the role of communication and linguistic abstraction in persuasion. First, our results deepen our understanding of how and why gender disparities in entrepreneurial funding outcomes might exist. The studies we present here suggest that some of these disparities may be driven not only by the deliberate bias that many scholars have suggested (e.g., Becker-Blease & Sohl, 2007; Greene, Brush, Hart, & Saparito, 2001) but may also be the result of a preference for entrepreneurs who communicate qualities that investors associate with venture success. Specifically, a gender disadvantage in entrepreneurship may be due to factors that extend beyond biased perceptions of "fit," whereby female entrepreneurs are seen as incongruent with entrepreneurship as a "masculine" endeavor that reveres traits such as aggressiveness and agency rather than warmth. Rather, investors filter and screen out investment opportunities based on an important objective, success-oriented criterion: the extent to which a company is scalable and has the potential for long-term growth. Abstract communication (which tends to correspond with how men generally communicate, rather than women) lends itself to investor attributions of a firm's orientation toward long-term growth and scalability. In this way, we find that investors make decisions based on criteria that they deem reasonable and essential, and it is only in the resulting investment outcomes that we observe a subsequent gender disparity.

Second, our findings also contribute to the larger body of literature on entrepreneurial financing. Grounding our work in studies on communication style allowed us to provide an important glimpse into the nuanced manner in which investors weigh and evaluate their decision criteria and the salience of communication style in the investment process. By demonstrating the impact of an abstract communication style in pitching one's message, we show that investment decisions often go beyond purely objective and economic considerations. Prior studies have shown that the inclusion of specific types of business data—such as target market size, formal founder credentials, competitor analyses, and product specifications (e.g., Fried & Hisrich, 1994; Wu, 2016), among others—can affect the likelihood that investors perceive an entrepreneur's venture as having long-term viability (Busenitz et al., 2005). Our work contributes to the recognition that even with the inclusion of this information, the style through which these factors are communicated can play an important role. Indeed, investors operate under conditions of extreme uncertainty, and the decisions that they make about how to allocate financial resources are often influenced by subtle signals and cues that they derive from how entrepreneurs present themselves. We show how communicating one's venture using a more abstract (versus concrete) communication style provides, particularly for women, one such signal about long-term viability—which is among the most critical investor considerations.

Finally, our findings offer novel contributions to the growing body of research on abstraction in cognition and communication. Research on construal level theory (Trope & Liberman, 2010) has primarily explored the impact of psychological distance on mental construal, meaning the degree to which people cognitively represent an object more abstractly or concretely. Recent work has extended this research to an interpersonal domain, exploring how audience characteristics—such as audience distance and audience size—affect communicative abstraction (Amit, Wakslak, & Trope, 2012; Joshi & Wakslak, 2014; Joshi et al., 2016). We further advance this research by showing how different groups of people may differ in their communicative abstraction in consistent ways, and critically, the important potential downstream consequences that this may have. In this way, our work provides an additional basis for understanding the complex interplay between communication and organizational outcomes and the ways in which specific means of communication, such as level of abstraction or construal levels, can affect individuals and founders in the entrepreneurship ecosystem.

Practical Implications

Our findings also contribute practical insights on the role of communication in entrepreneurial outcomes. Specifically, we provide evidence that both male and female entrepreneurs alike should recognize the role of communication style in investor evaluations—particularly, how communication goes beyond gendered stereotypes. Our findings demonstrate that although there may be disparities in entrepreneurial investing based on perceptions of one's orientation toward long-term growth and scalability, it is by this same token that there may be actions that entrepreneurs can take to combat such perceptions. Women may have the propensity to rely on concrete language, but by recognizing that they need to consider how communicative abstraction impacts perceptions of their long-term orientation and the importance of this perception for investment funding, they can more fully empower themselves. Furthermore, our exploratory findings in both Study 2 and Study 3 suggested that the link between perceptions of long-term orientation and investment decisions may operate differentially for women, as compared with men, suggesting that women should be conscious of not only whether they are effectively communicating this orientation via abstraction but also other associated factors that might impact perceptions of their long-term growth orientation and investment worthiness.

Limitations and Avenues for Future Research

We acknowledge a number of limitations in our research. First, our findings should be viewed in the context of a complex set of motivations and considerations that different types of investors take into account when making their investment decisions. Although we focused specifically on venture-backed firms that specifically seek external capital to grow and scale their startups (and in turn, investors who ultimately consider large financial returns as their measure of success), we acknowledge that other motivations could play a role in the signals that investors ultimately consider in their decisions.

Future research might expand on the current investigation in several ways. For example, as our exploratory analysis in Study 3 suggested, the impact

of communication style may be particularly nuanced for female entrepreneurs. Because female entrepreneurs are more likely to communicate concretely, they may be particularly penalized for not being seen as the type of long-term, big-picture thinkers that most align with investors' likelihood of investing. Additional work that examines the particular implications of communication style for female entrepreneurs—who may already face greater hurdles in their efforts to forge connections and establish homophily with investors (e.g., Huang & Mehta, 2018) and in their attempts to present themselves as embodying the image of a successful entrepreneur with long-term viability—would be useful to shed additional light on these findings.

It is also important that future work examines instances in which speaking concretely might not necessarily be a liability for male and female entrepreneurs alike. While we posit that communicative abstraction is relevant for perceptions of long-term growth and scalability in a startup, investors evaluating startups that have a strong social mission (e.g., Wu & Huang, 2017), for example, may show a different pattern than the one we observed here. To the extent that venture capital and equity investors desire to back long-term growth-oriented businesses, while microfinance institutions and organizations with a social mission might desire other types of business orientations, scholars may identify distinct ways that communication style affects decision processes and criteria.

Similarly, our work focused on entrepreneurial investment decisions and the allocation of financial capital, and there are a number of resources that are also critical for early-stage entrepreneurs beyond financial capital. For example, research has highlighted the benefits of concrete over abstract speech in both feedback (Kernan & Lord, 2018; Liden & Mitchell, 1985) and goal-setting contexts (Gollwitzer & Sheeran, 2006). As such, future research that examines how entrepreneurs seek advice, introductions, and other forms of human and intellectual capital could help provide more specific recommendations about whether communication style also plays a role and the extent to which individuals should strive to speak more abstractly, or the conditions in which speaking concretely might instead be warranted.

Finally, scholars may also consider how entrepreneurs might modify their communication style to include more abstract speech, and hence more directly manage the impressions of investors, for those situations in which abstract speech is indeed a factor in obtaining optimal outcomes. For example, it would be intriguing to see future work that explores the implications of different forms of modification of speech abstraction, e.g., whether it matters how one achieves abstraction or just that one does so.

CONCLUSION

Male entrepreneurs tend to communicate more abstractly than female entrepreneurs, and these gender differences in communicative abstraction result in differences in the degree to which investors perceive a venture to be oriented toward long-term growth and scalability. In turn, such investor perceptions affect how investors make investment decisions. We hope that this work spurs continued research on how individual differences affect communication styles, the role of gender in entrepreneurial evaluations, and how communication and other signaling mechanisms play a role in the success of both men and women in organizations more broadly.

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APPENDIX A

TABLE A1
Representative Examples of Abstract and Concrete Speech

Concrete		Abstract	
Focus on the "how" (or the feasibility of actions)	"WELL's tracking products include a wearable that uses transdermal technology to automatically track blood alcohol content. WELL's change products include a premium app with daily wellness notes and real-time coaching alerts, as well as web-based coaching for users who are really struggling to align their actions with their intentions."	Focus on the "why" (or the desirability of actions)	"The goal of LeaseCloud is to develop a fast solution for small and mid-sized companies to address the coming changes in accounting for leases expected to occur over the next four to five years. First, a little background—current lease accounting rules were developed over 30 years ago and enabled lessees to take most of their leases off the balance sheet. These rules have been criticized for many years by users of financial statements because they don't reflect the true financial position of companies. A recent exposure draft from the Accounting Standards Bodies ceases to address this by requiring lessees to capitalize their leases. In other words, bring them on the balance sheet."
	"GoNoGo is a free mobile app that offers a one-stop solution for most baby health needs. Our mission is twofold—we connect moms with qualified providers to help them decide whether or not to go to the ER. And second, we enable moms to get advice on their baby's health needs."		"There is currently a white space in the market because all services, from LinkedIn to local career counseling centers, help you to find just the next job while completely ignoring your long-term ambitions. With hard data and analytics as its backbone, ambitioni is ready to capture this white space."
	"We have a vision of fresh greens, local eggs, grass-fed beef . . ."		"We have a vision for healthy, farm-to-table food, available in every metropolitan area . . ."
Focus on details	"Most social drinkers can identify with the experience of waking up after a night out and wishing they had had at least one less drink the night before. They may have a hangover, they may have blown their diet, they may not remember aspects of the night. But they are not addicted; they don't want to quit drinking but they wish they had the tools to find the line between enjoying drinking and waking up feeling good. WELL gives users those tools."	Focus on essential elements	"In the past, the major accounting software vendors had been slow to address any kind of new accounting change. Especially for small to mid-size companies."
	"Our app offers an array of services that include Stella Health, local communities for moms, a repository of curated answers to frequently asked questions, ER and follow-up visit scheduling with doctors, and a date, a data platform to store baby health data safely and securely."		"We are an intuitive platform that helps students and professionals to do two things. One, explore potential long-term career plans. And two, connect with the relevant people behind those paths and learn from their experiences."
More verbs	"WELL is **moving** quickly to launch in 2014. In the near term, we are **focused on negotiating** for technology partnership and **testing** market demand. We **plan to spend** the summer in product development in order to **pilot** the program at universities in the fall and **iterate** for a holiday launch."	More adjectives	"For instance, when the **new** stock compensation rules were released about 10 years ago, **smaller** software vendors such as Equity Edge were more **successful** in establishing software solutions than **major** vendors such as Oracle and SAP. This is because these companies **better** addressed

TABLE A1
(Continued)

Concrete		Abstract	
			user needs and functionality and also analytics and reporting because they were specifically trying to address these **new** accounting changes, as opposed to incorporating a functionality into a **bigger** product"
	"Whenever a mom calls GoNoGo, she **gets connected** to an array of physicians who are **working** with these hospitals, and she can **see** the hospital she is **making** the call to."		"In today's information age, it is surprising that career planning is a very **difficult** process for most of us. Even at **reputed** schools like [School] and [School], students are pushed into **set** jobs, and there is no data about what lies ahead, even two years down the line. Professional networking is an **ineffective** process and knowledge discovery is limited to what your peers know."
Use of more words that can be experienced through the senses (concrete words)	Blood, alcohol, notes, night, drink	Use of words that are difficult to experience directly through senses	Capitalize, intuitive, successful, functionality, analytics
	Mom, baby, doctor, hospital, physicians		Knowledge, discovery, reputed, professional, learn

APPENDIX B

STUDY 3: EXPERIMENTAL STIMULI

In Study 3, investors were randomly assigned to one of four versions of a short description of an entrepreneurial venture: abstract–female, abstract–male, concrete–female, and concrete–male. A short passage of introductory text was shown to all conditions.

Text Shown to All Investors

The following excerpt is from a pitch that was given by the founder at a recent pitch competition. Put yourself in the role of an investor, imagining that you are considering investing in the opportunity. Then, answer the questions that follow, providing your assessment of the opportunity.

Text Shown to Investors Assigned to Abstract Style

Hi, I'm Jennifer [Jonathan] Martin, the founder of NANOSPRAY. Nanospray is a glass coating that is transparent, nontoxic, and can protect virtually any surface from damage. The Nanospray coating is highly flexible and breathable and can bond to a wide range of surfaces, protecting them from dirt, bacteria, UV light, and acids. The coating is nontoxic and environmentally safe. Nanospray uses path-breaking technological developments in nanotechnology to meet the market demands for a safe and easy-to-use surface maintenance product. Nanospray is the best new product of its type and has the potential to take the market by storm.

Text Shown to Investors Assigned to Concrete Style

Hi, I'm Jennifer [Jonathan] Martin, the founder of NANOSPRAY. Nanospray is a glass coating that is transparent, nontoxic, and can protect virtually any surface from damage. The Nanospray coating produces a water-resistant layer only around 100 nanometers (15–30 molecules) thick that can bond to any surface, including glass, wood, tile, brick, and stone. Any microbes landing on the sprayed surface cannot divide and replicate, and the nanoparticles in the spray repel dust and water while simultaneously protecting the surface from UV light and acids. The coating is made from natural materials and is safe for you as well as the environment.